

**NASA EARTH SCIENCE DIVISION
A.7 Water Quality Applications
NNH25ZDA001N-WATERQ**

Panel Review Summary Form

Proposal Number: 25-WATERQ25-0074
Proposal Title: Assessing Post-Storm Water Quality Patterns and Predictability for Recreation and Fisheries Management in the Coastal Carolinas

PROPOSAL CATEGORY:

☒ Type 1: Supports activities that are in the early stages of proving applications concepts (starting ARL 2-3)

☐ Type 2: Supports activities and products that aim to achieve operational deployment and sustained use in decision-making by the end of the award (starting ARL 5+)

Please note: Type 2 proposals are expected to have an ending ARL of 8-9 and are required to outline a Transition Plan as part of the Science-Technical-Management (S/T/M) Section.

BRIEF SUMMARY OF RESEARCH OBJECTIVES:

(2-6 sentences. In your own words, summarize the actual applied research or application proposed. Note the major decision-making need, objectives, methodology, and expected improvements to water management practices)

This project proposes applied research to improve North Carolina management of beach water quality advisories after storms and king tide events. The key decision-making need is the ability to issue and lift advisories more accurately and quickly, since current weekly sampling causes delays that can either risk public health or unnecessarily harm tourism. To address this, the team proposes to integrate satellite observations (e.g., turbidity, chlorophyll a, SST from NASA products such as PACE, GOES, VIIRS, perhaps GLIMR, as well as ESA products), high frequency beach sampling, and hydrodynamic-biogeochemical modeling to understand and predict patterns of degraded water quality, particularly enterococci. The methodology combines historical storm analyses, *in situ* measurements, and development of real time mapping tools in collaboration with NC Department of Environmental Quality (DEQ). Expected outcomes include improved timing and spatial resolution of water quality assessments, creation of decision-support maps ("nowcasts"), and more adaptive advisory practices that better protect public health while reducing unnecessary beach closures. The proposal progresses from Application Readiness Level (ARL) 3 to ARL 6 by transitioning from exploratory methods to practical applications tested and validated by our end-user partner at NC-DEQ.

PANEL CRITERIA FINDINGS:

INTRINSIC MERIT:

Evaluation of Intrinsic Merit includes the consideration of the following factors, as applicable to the particular proposal. Review panels are instructed not to compare proposals to each other; any comparative evaluations are conducted by NASA program personnel. In addition to the factors given in the ROSES Summary of Solicitation and the NASA Guidebook for Proposers, the evaluation criterion "Intrinsic Merit" specifically includes the following factors (See Section 5 in the A.7 solicitation):

**NASA EARTH SCIENCE DIVISION
A.7 Water Quality Applications
NNH25ZDA001N-WATERQ**

Panel Review Summary Form

- *The actual likelihood of successfully applying NASA Earth observations to make a valuable, substantive improvement with broad and scalable impact to a water quality topic of priority;*
- *Quality, creativity, and innovation of a project that advances the state of art in addressing water quality challenges;*
- *Intent, scope, and quality of strategy to engage end users, assess needs, and foster solution adoption, including a plan for accuracy assessment that demonstrates the application's ability to meet user requirements;*
- *Appropriateness of approach and methodology to effectively apply NASA Earth science data, model outputs, and simulated products, and evidence that the approach is likely to succeed;*
- *Feasibility of developing, testing (accuracy/uncertainty assessment), and implementing the proposed water quality application/solution;*
- *Likelihood of ARL advancement during the project period of performance; and*
- *Potential for tangible benefits to end users, improved outcomes in their actions from using Earth observation in their decision-making process, and/or measurable improvement in the lives and livelihoods that their decisions affect.*

Additional Intrinsic Merit considerations for Type 2 proposals:

- *Quality of the Transition Plan to integrate project results into end-user operations, addressing risks and enabling sustainable data solutions;*
- *Viability of pathways for sustained operations, including plans for long-term data management, evolving datasets, data latency, data volumes, new data algorithms/models, and cost–benefit justifications; and*
- *Likelihood and feasibility that the project applications will achieve operational deployment and sustained used in decision-making and will lead to concrete water quality actions by the end of the award.*

Intrinsic Merit Major Strengths:

The proposal presents a strong co-developed partnership with NC DEQ, featuring early and continuous engagement through quarterly check-ins, workshops, tool testing, and integration within existing advisory workflows, aimed at improving the speed and accuracy of issuing and lifting beach advisories. It addresses a clear and important management need by providing higher frequency, spatially detailed “nowcast” information on salinity, turbidity, chlorophyll-a, and SST to guide precautionary advisories and targeted sampling, especially under complex storm and king-tide conditions. The project is technically robust, integrating multiple NASA and ESA satellite missions including products from PACE OCI, VIIRS, GOES ABI, Sentinel-3 OLCI, future GLIMR, historical advisory analyses, targeted high-frequency beach sampling, dye-tracer studies, and additional optical sensors on existing buoys—to capture storm-driven dynamics and inland-to-coastal pollution transport at relevant scales. Its well-designed field component, including daily sampling during storm and king-tide events and state-approved Enterococci analyses, is a practical approach to quantifying how long degraded conditions persist beyond the

**NASA EARTH SCIENCE DIVISION
A.7 Water Quality Applications
NNH25ZDA001N-WATERQ**

Panel Review Summary Form

current weekly sampling cadence. With plans for regionally tuned algorithms, *in situ* validation, and a synthesis of stakeholder input through interviews and focus groups, the proposal establishes a pathway to an operational decision-support tool and provides a framework for replication in other U.S. coastal systems.

Intrinsic Merit Minor Strengths:

The proposal demonstrates a comprehensive field strategy, highlighted by dye-tracer releases, deployment of additional optical sensors on existing buoys, and daily sampling during storm and king-tide events, all of which will offer valuable observational context for understanding plume timing and water-quality changes. By leveraging established observing infrastructure including NC DEQ's long-term sampling records, CORMP buoys, gliders, and other regional assets, the project builds on a strong foundation rather than duplicating effort. It remains aligned with manager priorities by explicitly focusing on requested variables such as chlorophyll-a, turbidity, salinity, and SST, ensuring the EO elements stay relevant to decision needs. The structured end-user evaluation plan via using workshops, focus groups, cross-state interviews, and a final review, provides a clear mechanism for iterative feedback and verification that products meet management requirements. Deliverables are designed for practical adoption through integration with NC DEQ's existing web-based advisory platform, reducing barriers to operational transition. Finally, the open science and data stewardship plan, including SeaBASS archiving, code release, and DOI-assigned repositories supports transparency, reusability, and scalability to other regions and applications.

Intrinsic Merit Major Weaknesses:

The proposal's approach carries a few important weaknesses and uncertainties, beginning with its reliance on satellite-derived salinity, turbidity, and chlorophyll-a as imperfect proxies for bacterial risk, without clearly defined quantitative performance targets to guide precautionary or lifting decisions. Secondly, while validation with CORMP sensors and beach samples is mentioned, the absence of explicit acceptance criteria, error budgets, or operational skill metrics leaves unclear whether the products can truly "meet user requirements" in a conservative public-health context. Third, the proposed coupled hydrodynamic-sediment-biogeochemical modeling framework also raises technical concerns, as it aligns more naturally with physical and optical variables than with Enterococci dynamics. Similarly, the proposal references sediment reservoir mechanisms during king-tide events but does not fully explain how these processes will be quantified or translated into actionable advisory thresholds. Nearshore EO limitations, including land adjacency effects, optical complexity, and reduced coverage during storms and king-tide periods further challenge the feasibility of reliable beach-level mapping when it is most needed. More broadly, the end-to-end decision pathway is underdeveloped: although the management need is well framed, the specific advisory actions NC DEQ could defensibly take based on model and satellite proxies remain uncertain, and the proposal does not fully address false-positive and false-negative risks or define performance thresholds that would justify

**NASA EARTH SCIENCE DIVISION
A.7 Water Quality Applications
NNH25ZDA001N-WATERQ**

Panel Review Summary Form

changes in output. Additional concerns include the complexity of the multi-sensor processing and coupled modeling, which may exceed what state agencies can sustain without a clear hosting, maintenance, and training plan.

Intrinsic Merit Minor Weaknesses:

Although the proposal identifies its starting point as ARL 3, it lacks the foundational science required to justify that maturity level and appears more appropriately positioned at ARL 1. This gap is compounded by limited methodological detail on how nearshore satellite retrieval challenge such as adjacency effects, optical complexity, and turbid-water biases will be handled, as well as an absence of a quantified contingency plan for cloud cover or persistent data gaps during storm periods. While the coupled hydrodynamic-biogeochemical framework is technically interesting, its added value for Enterococci-based beach advisories remains unclear relative to the more direct dye-tracer, buoy, and beach-sampling components, and the proposal does not sufficiently address the difficulty of representing sediment-associated bacterial storage and remobilization during king-tide events. The proposal also remains vague on how forecast or nowcast information would translate into defensible advisory actions in a conservative public-health environment, especially since managers would still need buffer time to account for uncertainty before lifting warnings. Broader regulatory or stakeholder pathways are missing as well, with NC DEQ lacking established rules for precautionary advisories and the proposal offering no mechanism to develop consensus thresholds. Overall, the effort does not present a clearly defined problem, lacks specificity in how satellite products and methods will be applied, and does not convincingly demonstrate that the investigative team has the necessary expertise to execute the work.

RELEVANCE:

The evaluation of Relevance (for both Type 1 and Type 2 projects) will include the following:

- *Overall intent to apply NASA Earth observations to make potentially valuable, substantive improvements with broad and scalable impact to a water quality topic of priority (See Section 2); and*
- *Inclusion of at least one funded Co-Investigator or unfunded Collaborator from a partner an end-user organization in the proposal that has a direct decision-making role in water quality management;*
- *Clear identification of the decision-making process of the end user, their authority to implement water quality management action based on the information gained from this project, and how they will use the product to direct this action; and*
- *Alignment with Scope of Solicited Work, the NASA Water Resources Program, and description of how the project advances the NASA Earth Science to Action strategy.*

**NASA EARTH SCIENCE DIVISION
A.7 Water Quality Applications
NNH25ZDA001N-WATERQ**

Panel Review Summary Form

Relevance Strengths:

The proposal strongly fulfills the solicitation's intent by using NASA Earth observations to reduce coastal health risks and unnecessary beach advisories, with NC DEQ fully embedded as an active, funded end-user organization with direct management authority. It clearly identifies the current advisory workflow and shows how new satellite products and model outputs could inform sampling timing, advisory lifting, and spatial targeting. The project aligns well with the NASA Water Resources Program and Earth Science to Action strategy by integrating multi-sensor EO data (PACE, VIIRS, GOES, GLIMR) with hydrodynamic models and real-time mapping tools, all centered on manager requested variables such as chlorophyll-a, turbidity, salinity, and SST. Relevance is further strengthened by its focus on inland-to-coastal pollution transport and its thoughtful effort to broaden management value beyond single sampling locations. Finally, the proposal demonstrates strong potential for broader impact through its explicit emphasis on transferability supported by interviews, focus groups, collaborative planning, and a replicable framework applicable to other coastal regions facing similar stormwater and contamination challenges.

Relevance Weaknesses:

The proposal is relevant to beach water quality management, but it does not clearly demonstrate how the products would meaningfully influence NC DEQ's advisory decisions. While the requested variables are manager informed, the link to the regulated enterococci standard is weak, and the proposal provides limited explanation of how EO and model-derived information would translate into specific, defensible advisory actions rather than general situational awareness. Reliance on nearshore satellite products adds uncertainty due to adjacency effects, optical complexity, and cloud cover, while the technically ambitious coupled hydrodynamic-biogeochemical framework is not clearly justified for the advisory task. The king-tide resuspension mechanism is motivated but lacks detail on how sediment-associated bacteria will be represented well enough to support decision-making. Finally, operational testing is concentrated in the Cape Fear region, and scaling up depends on indirect generalization rather than direct multi-state validation.

REASONABLE COST:

Evaluation of the cost reasonableness addresses whether the proposed level of effort (i.e., labor FTEs) and the proposed other direct costs (i.e., supplies, equipment, travel) are commensurate with those required to accomplish the goals of the investigation. Salary levels, fringe benefit rates, and overhead rates are not displayed and are not part of that evaluation. Do not assess cost weaknesses for proposals on the basis of missing salaries. Low cost, while desirable, does not offset the importance of reasonableness of the proposed budget. The comparison of the proposed cost to available funds is performed by NASA program personnel and is not part of the peer-review process.

**NASA EARTH SCIENCE DIVISION
A.7 Water Quality Applications
NNH25ZDA001N-WATERQ**

Panel Review Summary Form

Cost Strengths:

The proposed budget appears generally reasonable and aligned with the scope and complexity of the work, with labor effort and direct costs appropriately matched to activities such as EO data processing, coupled modeling, field sampling, end user engagement, and tool development. The proposal efficiently leverages existing data sources and platforms to keep costs focused on tasks, and travel and equipment requests seem justified. Overall, the costs appear thoughtfully constructed and largely sufficient for achieving the project's stated goals. Workshops, focus groups, and meetings with NC DEQ and other potential adopters have budgeted support, ensuring the application can transition toward actual use rather than remaining purely research driven.

Cost Weaknesses:

The budget raises some concerns because the majority is allocated to subawards that handle core data-analysis and modeling tasks, suggesting that the lead team may lack the full expertise needed to execute the project. Several technically intensive components such as integrating multiple EO products, tuning regional algorithms, running fine-scale coupled models, and conducting rigorous validation appear under-scoped relative to the effort required, creating risk that planned work exceeds available labor and resources.

SUMMARY OF OVERALL PANEL EVALUATION:

This proposal tackles a real coastal public health need with innovative use of NASA Earth observations, fieldwork, and modeling, and it is grounded in NC DEQ's advisory workflow. It offers strong observational design and a credible path toward operational transition, though the problem definition, end-user collaboration, and linkage to enterococci based decisions are not always clearly articulated. Key risks include uncertain predictive strength of remote-sensing proxies, reliance on GLIMR, and incomplete validation criteria. Overall, the project is creative and relevant but needs sharper justification of how each component directly improves beach-advisory decision making. The panel evaluated this as a Good proposal.

OVERALL RATING:

- ☐ Excellent
- ☐ Very Good / Excellent
- ☐ Very Good
- ☐ Good / Very good
- X Good
- ☐ Fair / Good
- ☐ Fair
- ☐ Poor / Fair
- ☐ Poor